

Table 1: PWAs vs traditional web apps and native mobile apps

Feature	PWAs	Traditional web apps	Native mobile apps
Offline functionality	Can work offline with cached content	Typically unusable offline	Full functionality may require an internet connection
Installability	Installable on home screen	Not installable	Installed from the app store
Push notifications	Can send push notifications	No push notifications	Can send push notifications
Performance	Prioritise fast loading times and smooth performance	Performance can vary	Generally optimised for a specific device
Platform agnostic	Work across different platforms and devices (web browser)	Platform-specific (web browser dependent)	Platform-specific (requires separate downloads)
Discovery	Discoverable through search engines (SEO)	Discoverable through search engines (SEO)	Primarily discovered through app stores
Installation friction	No app store download required	Require accessing the website through browser	Require download and installation from app store
Storage footprint	Generally smaller footprint	Can vary depending on the complexity	Typically larger footprint due to full app installation
Device integration	Limited access to some device features (GPS, camera)	Limited access to some device features	Full access to device features

- Building a single codebase for PWAs can potentially streamline development compared to creating separate native apps for different platforms.
- Updates to PWAs can be deployed instantly through the web, eliminating the app store approval process required for native apps.
- PWAs work across various devices and platforms, extending a developer's reach without needing to create platform-specific versions.

Where PWAs are most effective: Key industries and use cases

PWAs aren't a one-size-fits-all solution, but they excel in specific industries and use cases. Here's a breakdown of where PWAs truly take centre stage.

- **E-commerce:** Imagine a world where your favourite online store feels as smooth and responsive as a native app, even on a slow internet connection. PWAs make this a reality for e-commerce businesses. With lightning-fast loading times, offline browsing capabilities (perfect for product research on the go!), and push notifications for promotions

and abandoned carts, PWAs can significantly enhance the mobile shopping experience and drive sales.

- **Publishing:** News outlets, magazines, and other content publishers can leverage PWAs to ensure their content is always accessible, even when users are offline. This allows readers to download articles or browse saved content for later, creating a seamless and uninterrupted reading experience. Push notifications also keep users informed about the latest updates and stories.
- **Travel:** No more scrambling for a Wi-Fi signal to access your travel itinerary or manage bookings! PWAs empower travel companies to create apps that function flawlessly offline. Users can download their flight information, hotel reservations, and even local maps before their trip, ensuring a stress-free travel experience regardless of internet connectivity.
- **Financial services:** Security is paramount when it comes to financial apps. PWAs offer a secure way for banks and financial institutions to provide users with mobile access

to their accounts, payment options, and financial tools. Additionally, PWAs can leverage push notifications for transaction alerts and security verifications, enhancing user trust and convenience.

How to develop PWAs

Here is the step-by-step process of creating a PWA based on your website.

Step 1: Preparing the foundation: Audit and manifest

Before diving into code, assess your website's readiness for a PWA transformation.

- **Audit your website:** Use tools like Google Lighthouse to identify performance bottlenecks and areas for improvement. Ensure your website loads quickly and offers a smooth user experience on mobile devices.
- **Create a web app manifest (JSON):** This file acts as the blueprint for your PWA, providing essential information like name, icons, and functionalities. Here's a basic structure to get you started (replace placeholders with your website's details):